

All in One Inverter and Lithium Battery For Residential Solar

- ✓ Rated Capacity:5.12KWH/16KWh
- ✓ Single Cell Capacity:100/280/304/314Ah
- ✓ System Configuration:1P16S
- ✓ Rated Current:50A/150A
- ✓ Weight:Approx. 49Kg/135KG
- ✓ Dimensions:560*487*932mm
- ✓ Protection Level:IP54
- ✓ Display:System status information
- ✓ Photovoltaic Inverter Power:10KW
- ✓ Low Voltage Switch:Yes
- ✓ DC Switch:Yes
- ✓ PV Input:PV1/PV2
- ✓ AC Input:AC220V/50HZ
- ✓ AC Output:AC220V/50HZ
- ✓ Accessories Base:Yes
- ✓ Wall Mounting Kit:Yes

Product Introduction

The CTECHI All-in-One Residential Energy Storage System (ESS) combines cutting-edge LiFePO4 battery technology with a powerful off-grid or hybrid inverter, offering an efficient and compact energy storage solution for modern homes. Designed for seamless integration with off-grid or on-grid and off-grid applications, this all-in-one system simplifies energy management by connecting directly to existing PV panels, mains power, and even diesel generators. It provides a reliable and eco-friendly way to store excess solar power during the day and deliver uninterrupted energy during peak hours or outages.

Superior Performance and Longevity

Equipped with premium LiFePO4 battery technology, the system delivers over 6000 cycles, ensuring durability and reliable performance for years. UPS-level switching within $\leq 10\text{ms}$, it provides uninterrupted power for critical loads.

Compact All-in-One Design

This integrated solution combines the inverter and battery, eliminating the need for complex installations. It supports direct connection to PV panels, the grid, and additional power

sources, offering flexibility and ease of setup for residential energy storage systems.

Advanced Technology and Scalability

Featuring patented LFP module technology, the system ensures consistent performance and enhanced safety. It supports parallel configuration of up to 8 systems, enabling scalability to meet larger energy demands.

Smart Connectivity and Easy Configuration

The system is equipped with WiFi-enabled smart management, allowing users to monitor and control energy usage via a dedicated app. This ensures energy efficiency and convenience at your fingertips.

Technical Data

If you have any questions during the browsing process, please feel free to contact us and communicate with us at any time.

Product Model	PT2025	PT3570S/PT3580S	PT4075/PT4080	PT5510S	PT5515SP/PT5516SP	PT6215/PT6216
Battery Type	LFP,3.2V/100AH	LFP,3.2V/280AH/314AH	LFP,3.2V/280AH/314AH	LFP,3.2V/200AH	LFP,3.2V/280AH/314AH	LFP,3.2V/280AH/314AH
Rated Energy	2.56kWH	7.168kWH/8.0384kWh	7.168kWH/8.0384kWh	10.240KWH	14.336kWH/16.0768kWh	14.336kWH/16.0768kWh
Rated Battery Voltage	25.6V	25.6V	25.6V	51.2V	51.2V	51.2V
Voltage Range	21.6V~29.2V	21.6V~29.2V	21.6V~29.2V	43.2V~58.4V	43.2V~58.4V	43.2V~58.4V
Depth of discharge	90%	90%	90%	90%	90%	90%
Rated Output Power	1800W	3500W	4000W	5500W	5500W	6200W
Max.Peak Power	3600VA	6000VA	8000VA	11000VA	11000VA	12400VA

Rated Output Voltage	230Vac	230Vac	230Vac	230Vac	230Vac	230Vac
Rated AC Frequency	50/60Hz	50/60Hz	50/60Hz	50/60Hz	50/60Hz	50/60Hz
Switch Time	10ms(typical)	10ms(typical)	10ms(typical)	10ms(typical)	10ms(typical)	10ms(typical)
Num. of MPPT Trackers	1	1	1	1	1	1
Max.PV Array Power	300W	4000W	6500W	6000W	6000W	8500W
Input Current	15A	13A	27A	22A	22A	27A
Max.Voltage of Open Circuit		500Vdc	500Vdc	500Vdc	500Vdc	500Vdc
MPPT Voltage Range	11~60Vdc	120~450Vdc	300~400Vdc	120~450Vdc	120~450Vdc	120~450Vdc
Input Voltage Range		90~280Vac	220~240Vac	90~280Vac	90~280Vac	360~430Vac
Frequency Range		50/60Hz	50/60Hz	50/60Hz	50/60Hz	50/60Hz

Bypass Overload Current		30A	6000W	40A	40A	9300W
Size	530*326*172mm	830*400*235mm	680*670*249mm	830*500*235mm	970*860*250mm	920*670*249
Weight	19kg	80kg	80kg	115KG	135kg	140kg
Protection Degree	IP20, Indoor OnIy	IP20, Indoor OnIy	IP20, Indoor OnIy	IP20, Indoor OnIy	IP20, Indoor OnIy	IP20, Indoo
Operating Temperature Range	-10~55°C,>45°C Derating	-10~55°C,>45°C Derating	-10~55°C,>45°C Derating	-10~55°C,>45°C Derating	-10~55°C,>45°C Derating	-10~55°C,>45°C Derating
Noise	<60dB	<60dB	<50dB	<60dB	<60dB	<50dB
Cooling method	Built-in fan	Built-in fan	Built-in fan	Built-in fan	Built-in fan	Built-in fan
Communication interface	RS485	RS485/CAN	RS485/CAN	RS485/CAN	RS485/CAN	RS485/CAN
External module (optional)		Wi-Fi / GPRS	Wi-Fi	Wi-Fi / GPRS	Wi-Fi / GPRS	Wi-Fi

Remark		Supports parallel battery expansion (customizable)	Supports both parallel power and battery expansion (customizable)	Supports both parallel power and battery expansion (customizable)	Supports both parallel power and battery expansion (customizable)	Supports both parallel power and battery expansion (customizable)
--------	--	--	---	---	---	---